enough to represent quantum states. For that, we need bras and kets, which have complex-valued components.

What sort of vector is the spin operator σ? It is definitely *not* a state-vector (a bra or a ket). It's not exactly a 3-vector either, but it does have a strong family resemblance because it's associated with a direction in space. In fact, we will frequently use σ as though it were a simple 3-vector. However, we'll try to keep things straight by calling σ a *3-vector operator*.

But what does that actually mean? In physical terms, it means this: Just as a spin-measuring apparatus can only *answer questions* about a spin's orientation in a specific direction, a spin operator can only provide *information* about the spin component in a specific direction. To physically measure spin in a different direction, we need to rotate the apparatus to point in the new direction. The same idea applies to the spin operator—if we want it to tell us about the spin component in a new direction, it too must be "rotated," but this kind of rotation is accomplished mathematically. The bottom line is that there is a spin operator for each direction in which the apparatus can be oriented.

3.4 CONSTRUCTING SPIN OPERATORS

Now, let's work out the details of spin operators. The first goal is to construct operators to represent the components of spin, σ_x, σ_y, and σ_z. Then we'll build on those results to construct an operator that represents a spin component in any direction. As usual, we begin with σ_z. We know that σ_z has definite, unambiguous values for the states $|u\rangle$ and $|d\rangle$, and that the corresponding measurement values are $\sigma_z = +1$ and $\sigma_z = -1$. Here is what the first three principles tell us:

- Principle 1: Each component of σ is represented by a linear operator.

- Principle 2: The eigenvectors of σ_z are $|u\rangle$ and $|d\rangle$. The corresponding eigenvalues are +1 and −1. We can express this